

JOSEPH DENG

Year 2, Computer Science & Physics Major

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TECHNICAL SKILLS

Programming: Python, Java, Racket, C

Tools: GitHub, Vercel, Contentful, Flask, Selenium, SQLite, Blender

Testing : JUnit, Writing test plans

Web: HTML5, CSS, JavaScript, TypeScript, Node.js, React.js, Next.js Bootstrap, Tailwind

TECHNICAL EXPERIENCE

Full-stack Developer | CS Toolkit (Startup)

May – June 2024

- Implemented dynamic API calls to Contentful using Flask to add pagination, tagging, and a search function to the company's blog
- Worked with a UX designer to reskin the blog to match a Figma design and went through a multi-stage build and review process to ensure that every detail was accounted for.
- Followed industry best practices for Git branching and pull requests

TECHNICAL PROJECTS

AI Recipe Generator

May – July 2024

- Implemented a Node.js backend that made asynchronous calls to the OpenAI API to create AI-generated recipes based on ingredients entered by the user.
- Designed the front-end web application with React to dynamically change visibility of elements, and Three.js to insert a custom interactive 3D model created using Blender.
- Refactored project to use the Next.js app router to deploy website using Vercel

Musical Ear Trainer

Jan – Apr 2024

- Built an interface with Java Swing that quizzes the user on randomly generated notes which are then played out loud using the MidiSynth package
- Encapsulated common data structures into classes to be easily instantiated and accessed
- Created a logical class hierarchy for storing previous quiz results which allowed the user to write and read their data to and from JSON.
- Designed a comprehensive testing suite using JUnit to ensure that the code is bug free

Web Scraper for Rate Your Music

Dec 2022- Jan 2023

- Utilized Selenium WebDriver to search for a particular artist then load and navigate HTML to find the items with the highest ratings and displayed the results using Tkinter

WORK EXPERIENCE

Program Assistant at Laurentian University | Shad Canada

June – July 2024

- Lead a group of 63 high-achieving students in grades 10 and 11 through a month-long STEAM and Entrepreneurship program.
- Presented an interactive beginner friendly workshop on 3D modelling using Blender.

EDUCATION

BSc. Computer Science & Physics | 4.0 GPA | *University of British Columbia*

2023 – 2028